

Can RAG-Driven Enhancements Amplify Audio LLMs for Low-Resource Languages?

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Abstract—The proliferation of Large Language Models (LLMs) has transformed Natural Language Processing (NLP), yet their development has largely overlooked low-resource languages. This paper addresses this disparity by evaluating three prominent Large Audio Language Models (LALMs) – LTU-AS, GAMA, and Pengi – across tasks like Automatic Speech Recognition (ASR), Audio Question Answering (AQA), and audio classification tasks in Hindi and code-mixed Hindi-English (aka Hinglish). We also explore the potential of Retrieval-Augmented Generation (RAG) to boost LALM performance in these low-resource settings. Our findings highlight significant performance discrepancies, with LALMs performing well in audio classification but struggling with ASR and AQA. While RAG shows potential, especially for audio classification, its impact is inconsistent across tasks. This work offers critical insights into the challenges of using LALMs for low-resource languages and provides a foundation for developing more inclusive and adaptable AI systems for complex multilingual tasks.

Index Terms—LLMs, RAG, AQA, ASR, LALMs

I. INTRODUCTION

Large Audio Language Models (LALMs) have demonstrated an extraordinary capacity to comprehend and deliberate upon our multifaceted world through language [1]. Traditionally, spoken-language understanding tasks, particularly Automatic Speech Recognition (ASR), have benefited significantly from the linguistic acumen of Large Language Models (LLMs) [9]. These models are increasingly being harnessed in an ever-widening spectrum of audio-centric applications, including Audio Question Answering (AQA) [8], audio event classification [7], and even audio captioning [3].

Contemporary multimodal LLMs often employ comparatively lightweight supervised fine-tuning strategies to align the model’s outputs with the user intent [8], [11], [13]. Recent audio-focused LLMs, illustrated by works such as [5] and [14], predominantly attend to the domain of speech. In contrast, other works have aimed to broaden the scope; LTU-AS [7] assimilates spoken text and speech paralinguistic cues, GAMA [6] enhances complex reasoning by integrating diverse audio representations, and Pengi [3] leverages transfer learning by reframing a wide variety of audio tasks as text-generation tasks. Yet, in spite of these advances, most extant models focus disproportionately on well-resourced languages, mainly English, thereby exposing significant gaps in both the

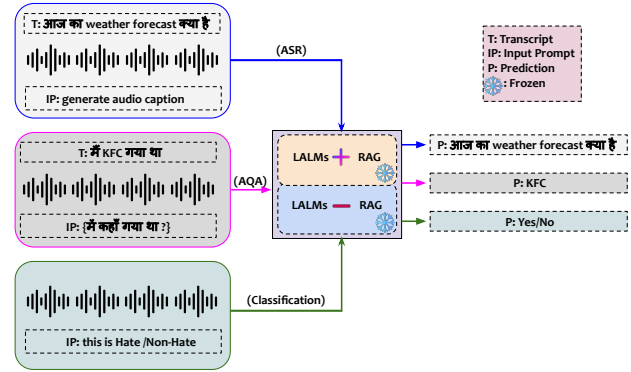


Fig. 1. The figure shows the framework used for evaluation in this study. We evaluate the LLM for three tasks (ASR, AQA, Audio classification.) The study can be used to quantify the gain by using RAG module in LLM.

development and thorough evaluation of models for under-represented linguistic communities. This disparity not only curtails the potential worldwide impact of LLMs but also signals a pressing issue of linguistic inclusivity in AI.

When we focus on low-resource languages such as Hindi and code-mixed Hindi-English (often termed as “Hinglish”), we encounter distinctive challenges. Their restricted training corpora, unique linguistic structures, and prevalent code-mixing render them far more complex for model building and assessment. ASR, audio classification, and AQA tasks become especially salient as they mirror real-world scenarios in multilingual contexts, demanding both lexical sensitivity and cultural nuance. Consider, for instance, a user querying a voice assistant in Hinglish: “Aaj ka weather forecast kya hai?” (“What is today’s weather forecast?”). To meet such a query appropriately, a sophisticated AI system must adeptly transcribe the utterance, interpret the intricate interplay of linguistic codes, and furnish a relevant response, thus emphasizing the intricacy of these undertakings in linguistically diverse environments.

The field of NLP has witnessed exponential interest in Retrieval-Augmented Generation (RAG) [10]—a paradigm that leverages external knowledge bases to enhance the performance of LLMs. While RAG has demonstrated advantages in various mainstream NLP applications, its implications for low-resource languages, particularly with regard to ASR, AQA, and classification tasks, remain unexplored. Addressing these

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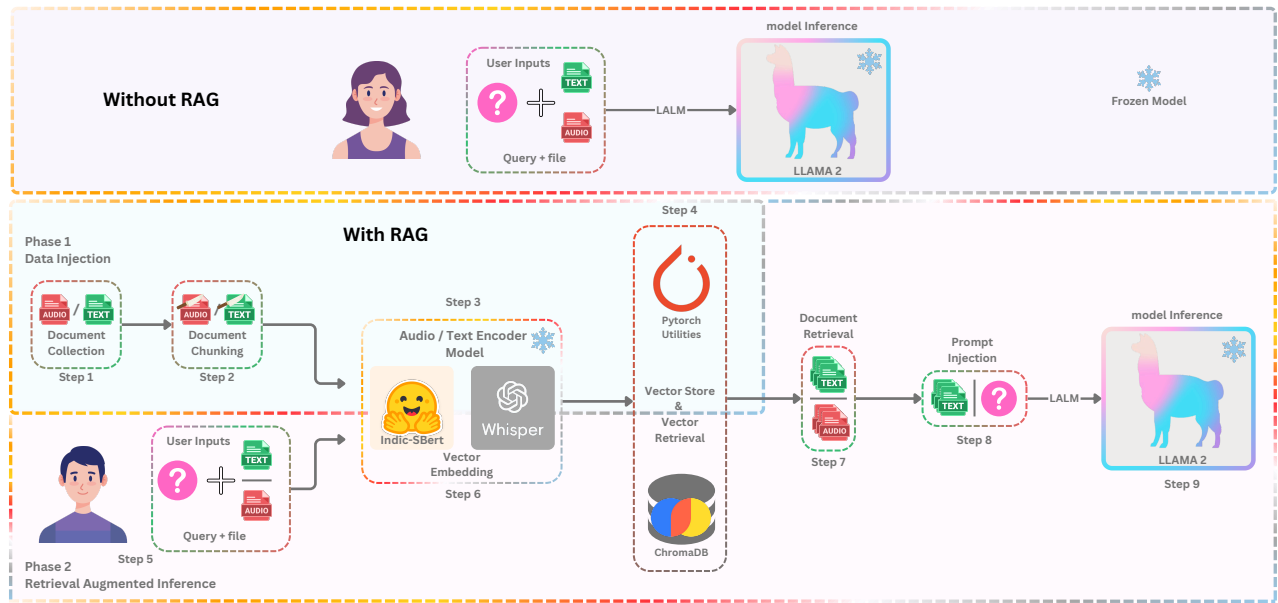


Fig. 2. Schematic comparison of Large Audio Language Model (LALM) inference with and without Retrieval-Augmented Generation (RAG). The RAG pipeline consists of two phases: (1) Data Injection, involving document processing and vector embedding, and (2) Retrieval Augmented Inference, encompassing query processing, semantic search, and prompt injection. Key components include PyTorch utilities, ChromaDB for vector storage and retrieval, audio/text encoders, and the LLAMA 2 model for final inference.

under-investigated avenues is crucial for extending the benefits of LLMs to a more diverse set of languages and applications.

In this work, we undertake a comprehensive evaluation of LALMs across three quintessential tasks (ASR, AQA, and audio classification) to assess their efficacy in low-resource linguistic environments, particularly those encompassing Indic languages. Specifically, we utilize three prominent audio LLMs: LTU-AS [7], GAMA [6], and Pengi [3], and examine their dexterity in coping with the linguistic and cultural complexities that permeate multilingual and code-mixed speech. While ASR assays the models' prowess in transcribing spoken content, AQA probes their capacity to interpret and respond to audio-based queries, and audio classification evaluates their competence in categorizing audio samples (including discerning hate speech). By juxtaposing performance with and without RAG, we emphasize the potential of retrieval-informed techniques to enhance model performance in under-served linguistic contexts. The objectives are twofold:

- 1) To evaluate the efficacy of state-of-the-art audio LLMs for ASR, Q&A, and audio classification tasks in low-resource, code-mixed language scenarios.
- 2) To elucidate the role of RAG in fortifying model performance across multilingual contexts, thereby laying the foundation for more equitable and universally relevant LLM solutions.

II. METHODOLOGIES

In this section, we present the details of our approach for evaluating three LALMs, namely Pengi [3], LTU-AS [7], and GAMA [6], across three key tasks: Audio Question Answering, Automatic Speech Recognition, and audio classification. Notably, LTU-AS is the only model trained on joint

audio-speech data. Our methodology proceeds in two phases: (1) a baseline evaluation of each LALM and (2) an enhanced approach that integrates RAG.

Baseline LALM Approach: In the baseline setup, we directly employ each model Pengi, LTU-AS, and GAMA without retrieval augmentation:

- **Model Inputs and Processing:** *Pengi* handles audio and text inputs simultaneously, enabling it to generate responses directly from the provided audio file and text prompt. *LTU-AS* and *GAMA* are more audio-centric, initially extracting features from the audio input, which are then combined with text prompts for final processing.
- **Task Outputs:** In *AQA* the models generate textual answers to questions regarding the audio content. In *ASR* the models transcribe the input audio into text. Finally, in *audio classification*, the models categorize the audio into predefined labels (e.g., hate speech detection).

In this phase, the models rely solely on their pre-trained parameters, providing a benchmark for subsequent comparisons with the RAG-enhanced approach.

RAG-Enhanced LALM Approach: We augment each LALM with a RAG pipeline to investigate the potential of retrieval-informed strategies. The process is as follows:

- 1) **Data Ingestion:** We treat each audio and text in our dataset as a distinct "context unit" and generate vector embeddings for these samples.
- 2) **Vector Embedding:** We employ Indic-sbert [2] for textual representations, given its demonstrated efficacy in low-resource environments. Further, we utilize Whisper [12] for audio representations, ensuring robust encoding of speech signals.

TABLE I
OVERVIEW OF THE DATASETS EMPLOYED FOR THE THREE KEY TASKS
EXAMINED IN THIS STUDY.

Dataset	Task	Metric	# samples
Indic QA	AQA	ACC, F1	1052
SLR 103	ASR	CER, WER	3843
SLR 104	ASR	CER, WER	3136
ADIMA	CLF	ACC, F1	3647

- 3) **Vector Storage and Retrieval:** The embeddings are stored in ChromaDB, an open-source vector database providing efficient similarity search.
- 4) **Input Processing:** During inference, query text and audio are converted to embeddings (via the same IndicSbert and Whisper models) to produce query vectors.
- 5) **Semantic Search:** We perform a cosine similarity-based search in ChromaDB to retrieve the top- k (where $k = 3$) context units, thus surfacing the most relevant information.
- 6) **Prompt Injection:** The retrieved contexts are appended to the original prompt via Alpaca-style short templates, ensuring compatibility with LALMs.
- 7) **Output Generation:** Augmented prompt is fed into the respective LALM, which synthesizes its response by leveraging the retrieved context and original input query.

Implementation Details: All three LALMs (Pengi, LTU-AS, and GAMA) are based on the Llama 2 backbone, inheriting its robust sequence modeling capabilities. As the tokens propagate through multiple transformer layers, the model progressively refines their context-aware representations, ultimately improving performance on AQA, ASR, and audio classification tasks. By comparing baseline outputs against those generated with RAG, we quantitatively assess how retrieval-based augmentation can enhance model efficacy in low-resource, multilingual settings.

III. EXPERIMENTAL SETUP AND RESEARCH QUESTIONS

In this section, we present our experimental protocol for evaluating LTU-AS [7], GAMA [6], and Pengi [3] on three core tasks: AQA, ASR, and audio classification. LTU-AS is the only model trained on joint audio-speech data. We also introduce the research questions (RQs) guiding our investigation and describe the datasets and evaluation metrics used.

Research Questions: We aim to address the following questions, both with and without the application of RAG:

- **RQ1:** What is the zero-shot ASR performance of LLMs on Hindi and code-mixed (Hindi-English) speech?
- **RQ2:** Can LLMs be effectively employed in zero-shot settings for audio classification tasks?
- **RQ3:** How do LLMs perform in zero-shot AQA on low-resource languages such as Hindi?

Datasets and Evaluation Metrics: For Audio Question Answering, we adopt the IndicQA [4] dataset, which supports multiple Indian languages. We focus on the Hindi subset, generating audio segments from the text-based content. Since the original passages in IndicQA are lengthy, we shorten them to

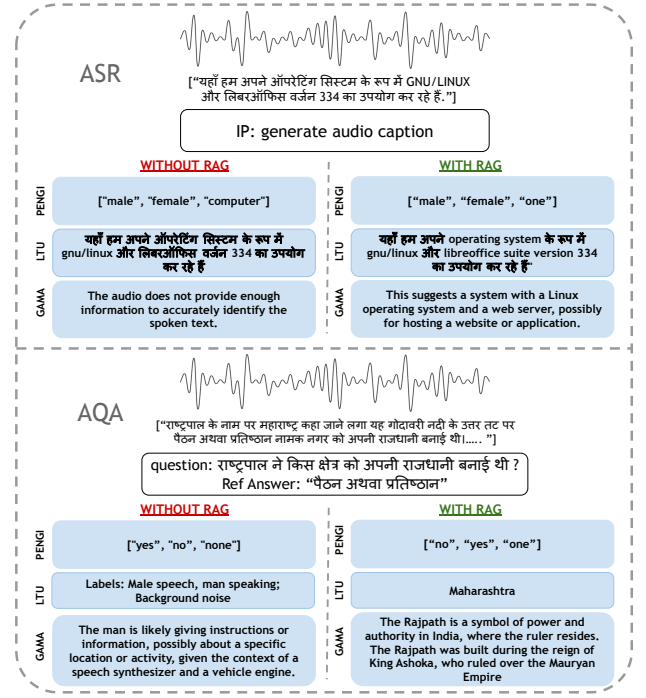


Fig. 3. Qualitative comparison of Large Audio Language Models (LALMs) performance on ASR and AQA tasks, with and without RAG

create an audio-oriented QA benchmark. Further, we employ SLR103¹ and SLR104² for ASR tasks, encompassing Hindi (SLR103) and code-mixed Hindi-English (SLR104) speech samples. Each dataset provides corresponding transcripts, enabling us to evaluate model performance via Character Error Rate (CER) and Word Error Rate (WER). For audio classification tasks, we use the ADIMA dataset, which spans 10 Indic languages and includes over 11,000 audio samples labelled as *Hate* or *Non-Hate*. We report accuracy and F1 scores to measure classification performance.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This section presents the empirical results for our three main tasks, with Tables II, III, and IV summarizing the key findings. Further, Figure 3 shows qualitative examples of ASR and AQA outputs from the models with and without RAG, illustrating how retrieval augmentation affects the actual transcription quality and response generation in practice.

A. ASR Performance (RQ1)

Baseline (No RAG). Among the evaluated models, LTU-AS exhibits the best performance on both SLR103 (Hindi) and SLR104 (code-mixed Hindi-English), with CER/WER of 0.646/0.604 on SLR103 and 0.812/0.821 on SLR104. Pengi delivers moderate results, while GAMA struggles noticeably, particularly on SLR103. Overall, performance on Hindi (SLR103) surpasses that on code-mixed data (SLR104), underscoring the added complexity posed by code-mixing.

¹<https://www.openslr.org/103/>

²<https://www.openslr.org/104/>

TABLE II

PERFORMANCE COMPARISON OF LARGE AUDIO LANGUAGE MODELS (LALMs) ON ASR TASKS FOR HINDI (SLR103) AND CODE-MIXED HINDI-ENGLISH (SLR104) DATASETS.

Models	LALMs - RAG				LALMs + RAG			
	SLR103		SLR104		SLR103		SLR104	
	CER	WER	CER	WER	CER	WER	CER	WER
LTU-AS	0.646	0.604	0.812	0.821	0.348	0.387	1.173	1.212
Gama	1.273	1.185	1.289	1.358	1.262	1.278	1.279	1.406
Pengi	0.946	1.000	0.907	1.001	0.956	1.014	0.939	1.032

GAMA’s weaker results hint at possible pre-training or architectural limitations for low-resource languages. In contrast, LTU-AS’s strong showing may stem from its joint audio and speech modelling, which appears advantageous in low-resource contexts.

With RAG. The impact of RAG varies across models and datasets. LTU-AS improves substantially on SLR103 (46.1% CER reduction, 35.9% WER reduction) but experiences degraded performance on SLR104. GAMA sees modest gains on both datasets, more pronounced for SLR103. LTU-AS’s improvement highlights RAG’s potential for enhancing monolingual ASR in low-resource scenarios, whereas the downturn on code-mixed data suggests challenges in retrieving suitable contexts. GAMA’s incremental yet consistent boost indicates some benefit from retrieval, although the model’s inherent limitations remain evident.

B. Audio Classification (RQ2)

Baseline (No RAG). Table III shows that, without RAG, LTU-AS achieves the highest accuracy (0.51), followed by Pengi (0.44) and then GAMA (0.38). LTU-AS attains the highest F1 score (0.62), indicating more balanced precision and recall. This outcome suggests that LTU-AS performs comparatively well across classes.

With RAG. Introducing RAG substantially boosts performance for LTU-AS and GAMA, while Pengi remains unchanged. LTU-AS’s accuracy leaps from 0.51 to 0.82, and its F1 score from 0.62 to 0.78, whereas GAMA’s accuracy rises from 0.38 to 0.87, with its F1 score more than doubling (0.22 to 0.82). Qualitative analysis reveals that the augmented models often provide more nuanced responses, effectively recognizing hate or non-hate content through contextual cues. Pengi’s stagnation may suggest an inability to leverage retrieved information for classification tasks.

C. Audio Question Answering (RQ3)

Baseline (No RAG). On the IndicQA dataset (Table IV), all models LTU-AS, GAMA, and Pengi perform extremely poorly in zero-shot conditions, with both accuracy and F1 scores at 0. These outcomes emphasize the formidable challenge of low-resource Hindi AQA, wherein errors can accumulate from both the ASR component and comprehension stage.

With RAG. Although LTU-AS and GAMA show slight improvements (accuracy up to 0.02 and 0.01, respectively), Pengi’s performance remains at 0. Despite these marginal gains, the results indicate significant difficulties for all models

TABLE III

ZERO-SHOT HATE SPEECH CLASSIFICATION PERFORMANCE OF LALMs ON THE ADIMA DATASET, COMPARING ACCURACY (ACC) AND F1 SCORES WITH AND WITHOUT RAG

Models	LALMs - RAG		LALMs + RAG	
	ACC	F1	ACC	F1
LTU-AS	0.51	0.62	0.82	0.78
Gama	0.38	0.22	0.87	0.82
Pengi	0.44	0.61	0.44	0.61

TABLE IV

ZERO-SHOT AUDIO QUESTION ANSWERING PERFORMANCE ON THE INDICQA DATASET FOR HINDI. RESULTS SHOW ACCURACY (ACC) AND F1 SCORES FOR LTU-AS, GAMA, AND PENGİ MODELS, BOTH WITHOUT (-RAG) AND WITH (+RAG) RETRIEVAL-AUGMENTED GENERATION.

Models	LALMs - RAG		LALMs + RAG	
	ACC	F1	ACC	F1
LTU-AS	0	0	0.02	0.04
Gama	0	0	0.01	0.01
Pengi	0	0	0	0

in handling Hindi-based AQA, even with retrieval augmentation. The poor outcomes can be attributed to limited exposure of the model to Hindi data and inherent challenge of accurately transcribing and interpreting non-English audio content.

Summary. These findings highlight the complexities of deploying LLMs for speech-centric tasks in low-resource settings. While RAG can offer notable gains in certain scenarios, particularly for monolingual ASR and audio classification its effectiveness varies considerably across tasks and language conditions. Further research on model architectures, targeted fine-tuning, and more robust retrieval mechanisms could mitigate these shortcomings and enhance performance on code-mixed and low-resource languages.

V. SUMMARY AND FUTURE WORK

This study offers critical insights into the capabilities and limitations of LALMs in low-resource settings, specifically focusing on Hindi and code-mixed Hindi-English scenarios. By evaluating LTU-AS, GAMA, and Pengi on ASR, AQA, and audio classification tasks, we uncovered the complex challenges these models face in code-mixed speech and complex query understanding. Our findings also highlight that while RAG can significantly boost performance—particularly in audio classification—its impact remains variable across tasks and models. Future research should prioritize strategies for handling code-mixed and multilingual data, including improved retrieval pipelines tailored to low-resource languages and specialized training paradigms that emphasize linguistic nuances. Developing comprehensive datasets and robust evaluation benchmarks for low-resource languages is equally crucial for advancing the inclusivity and applicability of LALMs.

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